

Capstone Project Solutions: Air Passengers

Section 0: The Data

0. Calendar Check (Code Cell)

```
# 0. Calendar check
n_obs = len(df)
range_ = (df["ds"].min(), df["ds"].max())
n_gaps = df["ds"].diff().gt(pd.Timedelta(days=31)).sum()
print(f"{n_obs} months, {range_[0].date()} to {range_[1].date()}, {n_gaps} gaps")
```

First Read Text (Markdown Cell)

The series shows a strong upward trend with an annual 12-month seasonal pattern whose variation expands as volume grows. There are no obvious cyclical fluctuations beyond the annual seasonal cycle.

Section 1: Exploration and Shape

1.1 Decompose (Code Cell)

```
from statsmodels.tsa.seasonal import STL

stl = STL(df["y"], period=SEASON_LENGTH, robust=True)
dcmp = stl.fit()

dcmp_df = df.copy()
dcmp_df["trend"] = dcmp.trend
dcmp_df["seasonal"] = dcmp.seasonal
dcmp_df["resid"] = dcmp.resid

P.decomposition_plot(dcmp_df, cols=["y", "trend", "seasonal", "resid"],
                    title="STL Decomposition: Trend and Seasonality Drive the Variance", x="ds")
plt.show()
```

1.2 Trend and Seasonal Strength (Code Cell)

```
var_resid = np.var(dcmp.resid)
var_trend_resid = np.var(dcmp.trend + dcmp.resid)
var_seas_resid = np.var(dcmp.seasonal + dcmp.resid)

trend_strength = max(0, 1 - (var_resid / var_trend_resid))
seasonal_strength = max(0, 1 - (var_resid / var_seas_resid))

print(f"trend strength: {trend_strength:.2f}")
print(f"seasonal strength: {seasonal_strength:.2f}")
```

1.3 Autocorrelation (Code Cell)

```
P.acf_plot(df["y"], nlags=24, title="ACF: Strong Seasonal Spikes at Lags 12 and 24")
plt.show()
```

Section 2: The Floor

2.1 What is left on the table (Code Cell)

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```
from statsmodels.stats.diagnostic import acorr_ljungbox

resid_floor = df["y"].diff(12).dropna()
lb_floor = acorr_ljungbox(resid_floor, lags=[12, 24], return_df=True)
print(lb_floor)
```

Section 3: Your Model

3.1 Read the components (Markdown Cell)

AutoETS selected the (M,A,M) variant (Multiplicative Error, Additive Trend, Multiplicative Seasonality). This matches our visual read from Section 1, where seasonal swings expand proportionally with the trend level.

3.2 The Forecast (Code Cell)

```
ets_fc = ets_sf.forecast(df=df, h=H_REPORT, level=scoring.LEVELS)

P.fan_chart(df, ets_fc.rename(columns={
    "AutoETS": "mean",
    "AutoETS-lo-20": "lo-20", "AutoETS-lo-40": "lo-40",
    "AutoETS-lo-60": "lo-60", "AutoETS-lo-80": "lo-80", "AutoETS-lo-95": "lo-95",
    "AutoETS-hi-20": "hi-20", "AutoETS-hi-40": "hi-40",
    "AutoETS-hi-60": "hi-60", "AutoETS-hi-80": "hi-80", "AutoETS-hi-95": "hi-95",
    "ds": "ds"}),
    title="AutoETS 24-Month Forecast with Prediction Intervals",
    history_tail=36, levels=(80, 95))

plt.show()
```

Section 4: The Harness

4.0 Score Cross-Validation (Code Cell)

```
RESULTS = scoring.score_cv(cv, levels=scoring.LEVELS)
RESULTS
```